

A generating grammar for six-point homometry in finite cyclic groups

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Abstract

Two subsets of a finite cyclic group are homometric when their unordered interval multisets agree. We give an explicit generating grammar for six-element subsets in every finite cyclic group: the connected components of its graph on translation/inversion classes are exactly the homometry classes. The grammar combines the classical two-parameter integer six-point construction, conditional block translations and reflections, a parallelogram-dyad exchange, half-per-coset complementation, autocorrelation-preserving units and thirteen rigid cyclic templates. The completeness argument uses universal signed edge-matchings. Their finite presentations reduce to source orders at most 135; their positive free-rank presentations are covered by a weighted real-line theorem and finite integral quotient certificates. The proof specifies the matching, height-arrangement and quotient algorithms and their acceptance conditions. The computations have separately implemented checks, but remain explicit trust dependencies. The largest two-method six-subset census is through modulus 135. The result concerns generation with compositions, rather than unique normal forms, minimal generators or closed counting formulas. No novelty claim is made.

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1 The problem and the claim

For a set of pitch classes, an interval inventory forgets where its intervals occur. This loss of placement is the mathematical question studied here; it does not assert equal harmonic function or identical recorded sound. In twelve-tone equal temperament, C major $\{0, 4, 7\}$ and C minor $\{0, 3, 7\}$ have the same interval classes $\{3, 4, 5\}$ and are connected by the reflection $p \mapsto 7 - p$. They belong to one translation/inversion class. The four-note sets $\{0, 1, 4, 6\}$ and $\{0, 1, 3, 7\}$ also have equal interval inventories, but occupy different such classes.

The four-point classification is prior work, including Rosenblatt’s theorem joint with Berman [1, Theorem 3.9]. Erickson and Jones give seven five-point types, including a triple family [2]. General constructions and finite enumeration also precede this work [4, 5]. The exact two-parameter six-point formula and its signed factorization appear as Family N in Yovanof–Golomb [7, p. 47]; the distinct-distance context of that paper is not used to infer weighted completeness here. The repeated-distance example at $(p, q) = (1, 4)$ is attributed to Hosemann and Bagchi (1954) in Lemke–Skiena–Smith [8, Table 1]. Classical block translations appear in the Callender–Hall handout [9, Cases 3–4], which credits Bullough for Cases 4a/4b. Goyette describes translation and reflection formulations [10, Formulas 3.2 and 3.5]; his proposed criteria are explicitly empirical, so they are not proof dependencies here. The symbolic-matching treatment of the six-point integer mechanism is also relevant [3]. Our objective is an explicit grammar valid at arbitrary modulus, rather than a finite census or an unrestricted spectral-unit existence statement.

The argument has two kinds of step. Mathematical lemmas prove soundness, normalization, exhaustive reductions, specialization and path inflation. Finite verification statements exhaust the resulting matching presentations and certify explicit mechanisms. We specify the latter algorithms and acceptance predicates as part of the proof, rather than treating reported totals as evidence of exhaustiveness. Section 11 identifies the saved inputs, arithmetic witnesses and independent checker scopes. The weighted real-line result also uses a precisely stated exact unsatisfiability obligation.

These computational dependencies are retained in the final conclusion; a bounded census alone would not prove it.

The statement is a generation theorem. It does not by itself solve the stronger classification-and-enumeration problem of choosing irredundant parameters and deriving counts. We distinguish that remaining problem from the completeness of the graph's generating grammar.

2 Definitions and the generating graph

Let $C_n = \mathbb{Z}/n\mathbb{Z}$, with n a positive integer. For a finite subset $A \subset C_n$, define its directed autocorrelation by

$$r_A(t) = \#\{(a, a') \in A^2 : a - a' = t\}.$$

For $1 \leq d \leq \lfloor n/2 \rfloor$, its interval-class count is the number of unordered pairs at cyclic distance d . Away from an antipodal distance, this count is $r_A(d)$; at $d = n/2$ it is $r_A(d)/2$. For sets of the same cardinality, equality of interval inventories is therefore equivalent to $r_A = r_B$. We call this relation homometry.

Translation and inversion act by $A \mapsto t + A$ and $A \mapsto t - A$. Their orbits are T/I classes. Two homometric sets in different T/I classes are a *Z-pair*. A nontrivial homometry family consists of at least two such classes. Multiplication by a unit is *not* included in the definition of a vertex.

In the integral group ring $\mathbb{Z}[C_n]$, write a set for its indicator, x^s for translation by s , and F^* for exponent reversal. Then homometry is $AA^* = BB^*$. This group-ring identity concerns pitch-class indicators; it is not an equality of audio spectra.

Definition 2.1 (Admissibility). *Every endpoint produced by a construction below must contain exactly six distinct elements. A named block partition must be disjoint. Independent translations, inversions and interchange of endpoints are permitted. When forming an edge, discard endpoints in the same T/I class.*

Let $\mathcal{G}_n$ have the T/I classes of six-subsets as vertices. Join two distinct vertices if representatives satisfy one of the explicit constructions in Section 3. Edges are undirected.

Theorem 2.2 (Six-point generation; Theorem G). *For every positive integer n , two six-subsets of C_n are homometric if and only if their vertices in $\mathcal{G}_n$ are joined by a finite path. Equivalently, the connected components are exactly the homometry classes.*

The assertion is vacuous for $n < 6$. Singleton components are allowed. The theorem permits compositions; it does not assert one direct move for every pair. It gives neither a disjoint parameter classification nor a minimal generating set. Its in-house computer-assisted status and external review boundary are specified in Section 12.

3 The explicit grammar

3.1 Four conditional block moves

Choose a disjoint partition $A = U + W$ with $|U| + |W| = 6$, and put $P = UW^*$. The new moving block must again be disjoint from U .

Move	Endpoint B	Required group-ring identity
L2: periodic translation	$U + x^s W$	$x^s P = P$
L3*: cosymmetric translation	$U + x^s W$	$x^{-s} P = P^*$
L4: halfturn translation	$U + x^s W$	$2s = 0, x^s(P + P^*) = P + P^*$
L5: block reflection	$U + x^c W^*$	$UW^* = x^{-c} UW$

These tests concern the small cross-difference distribution of the chosen blocks. L3* does not assume that s has a half in C_n ; replacing it by a halved-shift formula would lose legitimate cases.

3.2 B: the classical six-point family

For $p, q \in C_n$ define

$$X = \{0, p, q - 2p, 2q - 2p, 2q, 3q - p\}, \quad (1)$$

$$Y = \{0, p, q + 2p, 2q - p, 2q + p, 3q - p\}. \quad (2)$$

Require admissibility and different T/I classes. This formula and its signed factorization appear as Family N in Yovanof–Golomb [7, p. 47]. We retain the short name Bloom for this mechanism; neither the formula nor the factor identity is claimed as new. When both lists have six distinct integer points, integer p, q give line-homometric sets. Every admissible modular image has an integer lift of that kind.

3.3 D: a parallelogram-dyad exchange

Choose $a, b \in C_n$ with four distinct corners $R = \{0, a, b, a + b\}$, and let $\rho(t) = a + b - t$. Choose nonnegative integer functions U_0, V_0 that are constant on $\langle a \rangle$ -cosets and $\langle b \rangle$ -cosets, respectively, with total combined mass twelve. Put $K = U_0 + V_0$ and require

- $\rho(K) = K$ and $K = 1$ on R ;
- $K(t) \in \{0, 1, 2\}$ off R ;
- $K(t) \in \{0, 2\}$ at ρ -fixed points off R .

Choose a four-set C off R as follows. On a two-point ρ -orbit, choose neither point for $K = 0$, either point for $K = 1$, and both for $K = 2$. At a fixed point choose it precisely when $K = 2$. Set

$$X = C \cup \{0, a + b\}, \quad Y = C \cup \{a, b\}.$$

The mass condition is

$$12 = \text{ord}(a) \sum_{\langle a \rangle\text{-cosets}} U_0 + \text{ord}(b) \sum_{\langle b \rangle\text{-cosets}} V_0.$$

At least one step has order at most twelve. Nonunit steps are permitted. The periodic-weight decomposition is the reviewed cyclic kernel theorem D; the construction does not ask an unrestricted homometry oracle for a partner.

3.4 L7: half-per-coset complementation

Choose a subgroup H of even order $h \in \{2, 4, 6, 12\}$, and $12/h$ distinct H -cosets. In each select $h/2$ points. Let A be their union, let S be the union of the full selected cosets, and set $B = S \setminus A$. The $n = h = 12$ case is classical hexachord complementation.

3.5 L6: a conditional unit move

For a unit u modulo n , set $B = uA$ only when

$$r_A(ut) = r_A(t) \quad \text{for every } t \in C_n.$$

The condition is essential: an arbitrary unit need not preserve the interval vector. It is an actual edge between T/I vertices, rather than a silent change to their equivalence relation. L6 is used only in the bounded finite branch of the completeness proof.

3.6 R: thirteen rigid cyclic templates

For a row (q, X, Y) of Table 1, choose $t \in C_n$ with $qt = 0$, and produce tX, tY . The allowed orders are precisely those giving six-distinct and T/I-distinct endpoints. Subgroup inflation is included.

q	X	Y	$\text{ord}(t)$
17	0, 1, 2, 3, 8, 12	0, 1, 2, 6, 7, 9	17
19	0, 1, 2, 3, 6, 10	0, 1, 2, 4, 5, 11	19
21	0, 1, 2, 4, 7, 14	0, 1, 3, 7, 8, 10	21
21	0, 1, 2, 5, 6, 15	0, 1, 2, 6, 7, 10	21
21	0, 1, 3, 7, 10, 15	0, 1, 4, 7, 14, 16	21
23	0, 1, 2, 3, 7, 17	0, 1, 2, 4, 17, 18	23
24	0, 1, 2, 5, 7, 16	0, 1, 2, 6, 9, 11	12, 24
27	0, 1, 2, 3, 7, 19	0, 1, 2, 3, 8, 12	27
27	0, 1, 2, 6, 19, 22	0, 1, 3, 17, 21, 22	27
28	0, 1, 2, 4, 12, 23	0, 1, 3, 5, 11, 12	14, 28
30	0, 1, 2, 6, 19, 22	0, 1, 3, 9, 13, 14	15, 30
30	0, 1, 3, 5, 12, 25	0, 1, 6, 9, 11, 13	15, 30
31	0, 1, 2, 5, 11, 19	0, 1, 2, 6, 20, 23	31

Table 1: A sufficient rigid-template list; neither minimality nor independence from the other moves is asserted.

4 Soundness and specialization

Proposition 4.1. *Every admissible grammar edge preserves autocorrelation.*

Proof. For a block partition,

$$AA^* = UU^* + WW^* + P + P^*.$$

Translation of W replaces its cross sum by $x^{-s}P + x^sP^*$ while preserving WW^* . L2 leaves both cross terms unchanged; L3* exchanges them; L4 preserves their sum because $s = -s$. For L5 the first new cross term is $x^{-c}UW = P$, and its involution supplies the second.

For B use the Laurent polynomials

$$F = 1 + u + v, \quad Q = u^2 + uv^2 - uv + v, \quad R = u^2v - uv + u + v^2 = u^2v^2Q^*.$$

Direct expansion gives $X = u^{-2}FQ$, $Y = u^{-1}FR$ under $u = x^p, v = x^q$. Therefore $XX^* = YY^*$. The signed factor and its cancellations are essential. A collision-free 0/1 direct-sum flip with factor cardinalities two and three is rigid: every two-point factor is a translate of its reflection, so flipping either factor gives only a translation or reflection of the whole set.

For D, the selections have $K = C + x^{a+b}C^* + (1 + x^a)(1 + x^b)$. Its mass is $2|C| + 4 = 12$, so $|C| = 4$. Expansion gives

$$XX^* - YY^* = x^{-a-b}(1 - x^a)(1 - x^b)K. \quad (3)$$

The two periodic weights are annihilated by the displayed product. For L7, the half-coset condition gives $SA^* + AS^* = SS^*$, hence $(S-A)(S-A)^* = AA^*$. L6 permutes the prescribed equal coefficients. For R, counting the fifteen unsigned differences in each table row gives equality in C_q , which survives $x \mapsto t$. $\square$

Homomorphic transport of homometry has classical antecedents [6]. All group-ring identities transfer through homomorphisms. To use a transferred identity as a named generator, its combinatorial conditions must also survive. Injectivity on each endpoint preserves block disjointness. A halfturn that becomes zero gives an equal pair. For D, if the images of the equal-sum dyads share one corner, they share both, so the two endpoints coincide. Every nontrivial image retains four distinct corners, and the cyclic periodic-kernel result D supplies its periodic weights.

Lemma 4.2 (The L7 degeneration lemma). *Under a homomorphism injective on each endpoint of a six-point L7 pair, every T/I-distinct image is itself literal L7.*

Proof. If the kernel on H has size k , a half-coset's $h/2$ points must inject into h/k positions, so $k \leq 2$. For $k = 2$, both complementary halves fill the image coset; the two entire endpoints agree. Thus $k = 1$ in a nontrivial image.

At most two occupied source cosets can now merge, since each contributes $h/2$ distinct points to an image coset of size h . A merger gives one full common image coset. For $h = 2$, translation by the nonzero element of H exchanges every remaining singleton half and fixes every full coset; the whole pair is translation equivalent. For $h = 4$ there are three occupied cosets. A merger leaves one complementary dyad; each dyad in C_4 translates to its complement, with the same translation fixing the common full coset. For $h = 6$, merging exhausts the two cosets and makes the endpoints equal. For $h = 12$ there is only one coset. Thus a T/I-distinct image has no merger and retains literal L7. $\square$

5 The dyad kernel and complete shape parameterization

The periodic weights in D can be generated without solving an inverse homometry problem. We give the kernel theorem and the recovery argument explicitly. Throughout this section, functions have integer coefficients on C_n ; translation acts by $x^a W(t) = W(t - a)$.

Lemma 5.1 (Nonnegative periodic decomposition). *Let $W: C_n \rightarrow \mathbb{Z}_{\geq 0}$ and $a, b \in C_n$. Then*

$$(1 - x^a)(1 - x^b)W = 0 \iff W = U + V,$$

where U, V are nonnegative integer functions satisfying $x^a U = U$ and $x^b V = V$. If W has total mass m , then

$$m = \text{ord}(a) \sum_{C_n/\langle a \rangle} U + \text{ord}(b) \sum_{C_n/\langle b \rangle} V,$$

where the sums take one value per coset.

Proof. The reverse direction follows by annihilating the two summands. For the forward direction use the characters $\chi_k(t) = \exp(2\pi i k t/n)$, $0 \leq k < n$. Translation is diagonal on these characters, and a product eigenvalue $(1 - \chi_k(-a))(1 - \chi_k(-b))$ vanishes precisely when at least one of its factors does. Hence the complex kernel of the product is the sum of the two periodic subspaces. Taking real parts gives real periodic functions U_0, V_0 with $W = U_0 + V_0$.

This alone does not give nonnegative integral summands. Put $H = \langle a \rangle$, $J = \langle b \rangle$ and $L = H + J$, and work in one coset of L . List its H -cosets by i and its J -cosets by j . Every H -coset meets every J -coset: the difference between their representatives belongs to $H + J$. All intersections are nonempty, have the same size $|H \cap J|$, and partition the component. The function $W = U_0 + V_0$ is constant on an intersection; call its integer value w_{ij} . It satisfies the rectangle identities

$$w_{ij} - w_{ij_0} = w_{i_0j} - w_{i_0j_0} \quad \text{for all } i, j.$$

Choose j_0 and set $m_0 = \min_i w_{ij_0}$,

$$u_i = w_{ij_0} - m_0, \quad v_j = w_{i_0j} - w_{i_0j_0} + m_0.$$

These numbers are integral, $u_i \geq 0$, and the rectangle identities give $v_j = \min_i w_{ij} \geq 0$ and $w_{ij} = u_i + v_j$. Assign u_i to its whole H -coset and v_j to its whole J -coset. Repeat on each L -coset. The resulting U, V are the required nonnegative integral functions. Summing their constant values along cosets proves the mass formula. $\square$

Proposition 5.2 (Exactness of D within the dyad shape). *Suppose independent rigid motions align a six-pair to*

$$X = C \sqcup \{0, a + b\}, \quad Y = C \sqcup \{a, b\},$$

where $|C| = 4$ and $R = \{0, a, b, a + b\}$ consists of four distinct elements outside C . The pair is homometric if and only if it has the periodic weights and orbit selections specified in construction D.

Proof. Write $\rho(C) = x^{a+b}C^*$ and $K = C + \rho(C) + 1_R$. The expansion (3) and invertibility of x^{-a-b} show that homometry is equivalent to $(1 - x^a)(1 - x^b)K = 0$. The function K is nonnegative, integral, has mass twelve, and is fixed by ρ . Its coefficients are one on R and belong to $\{0, 1, 2\}$ off R . At a ρ -fixed point off R its coefficient is twice the coefficient of C , and therefore belongs to $\{0, 2\}$. Lemma 5.1 supplies exactly the required weights.

Conversely, admissible weights give a K with these properties. On each two-point ρ -orbit, choose no point, one point, or both points according as its coefficient is zero, one, or two. At a fixed point choose it precisely when its coefficient is two. This produces a binary set C off R with $C + \rho(C) = K - 1_R$. Summing coefficients gives $2|C| = 12 - 4 = 8$, so its cardinality is four. The dyad identity now gives homometry. The same selection rule recovers every C in the forward direction. The final T/I test decides whether it is a Z-pair. $\square$

If $\text{ord}(a) > 12$ then its periodic summand must vanish, because a nonzero nonnegative integer weight would alone contribute more than twelve. Thus K is b -periodic and $\text{ord}(b)$ divides twelve. If both orders exceed twelve, the construction is impossible. These are consequences of the exact mass formula, not search caps. This proposition classifies the displayed exchange shape; it does not assert that every six-pair can be aligned to that shape.

5.1 Coefficient proof of half-per-coset complementation

Let H have even order h , and let A contain $h/2$ points in each of $m = 12/h$ selected H -cosets. Write S for their full union. For any $t \in C_n$, let N_t count the ordered pairs of selected cosets (K, J) with $K = t + J$. Then direct coefficient counting gives

$$(SS^*)(t) = hN_t, \quad (AS^*)(t) = (SA^*)(t) = \frac{h}{2}N_t.$$

For example, for each such pair (K, J) exactly $h/2$ choices of a point of $A \cap K$ have their difference- t partner in J . Consequently $AS^* + SA^* = SS^*$, and expansion of $(S - A)(S - A)^*$ proves L7. The equality of the two cross coefficients here follows from the half-coset hypothesis, rather than from a general symmetry assumption on A . The number of selected cosets is integral precisely for $h \in \{2, 4, 6, 12\}$ in the six-point construction.

6 Universal signed matchings and the original presentation

We distinguish a relation in an integral presentation from a relation that holds only after a particular specialization. This distinction is essential: dividing an integral relation by its content can remove precisely the torsion that creates a cyclic $\mathbb{Z}$ -pair.

Lemma 6.1 (Occurrence matching). *Let $(a_i)_{i=0}^5$ and $(b_i)_{i=0}^5$ be two lists in an abelian group, including multiplicities. Their directed autocorrelations agree if and only if there are a bijection σ of their fifteen unordered edge occurrences and signs $\epsilon_{ij} \in \{1, -1\}$ such that*

$$a_j - a_i = \epsilon_{ij}(b_l - b_k), \quad (k, l) = \sigma(i, j), \quad i < j, k < l.$$

Proof. Each unordered edge with difference g contributes the two directed occurrences $g, -g$. Partition differences into orbits under $g \mapsto -g$. For an orbit of size two, its autocorrelation coefficient at either element is the number of unordered edges in that orbit. For a nonzero orbit of size one it is twice that number. At zero it is six plus twice the number of coincident off-diagonal unordered edges. Consequently equality of autocorrelations gives equal numbers of edge occurrences in every orbit. Match those occurrences bijectively within each orbit and choose a sign that gives equality. If $g = -g$, both signs are available and neither may be discarded. Conversely, a signed bijection equates the two directed occurrences contributed by each edge, and the six diagonal zeros agree. $\square$

Write $\mathcal{E} = \{(i, j) : 0 \leq i < j \leq 5\}$. In $\mathbb{Z}^{10}$ use the row vectors $e_1, \dots, e_5, f_1, \dots, f_5$ as the standard basis, and set $e_0 = f_0 = 0$. Independently translate each input endpoint so that $a_0 = b_0 = 0$. A signed matching defines the 15×10 matrix with rows

$$m_{ij} = e_j - e_i - \epsilon_{ij}(f_l - f_k), \quad (k, l) = \sigma(i, j). \quad (4)$$

The *integral relation lattice*, the *original presentation*, and its free rank are

$$L_M = \sum_{(i,j) \in \mathcal{E}} \mathbb{Z} m_{ij}, \quad G_M = \mathbb{Z}^{10} / L_M, \quad d = 10 - \text{rank}_{\mathbb{Q}} M. \quad (5)$$

We write E_i, F_i for the classes of e_i, f_i in G_M . The two labelled lists (E_i) and (F_i) are called the universal configurations.

Lemma 6.2 (Universal property and rank). *The universal configurations are homometric as lists. Every realization of the signed matching in an abelian group K factors through a homomorphism $G_M \rightarrow K$. Moreover $5 \leq \text{rank}_{\mathbb{Q}} M \leq 10$, so $0 \leq d \leq 5$. A realization by six-element sets rules out universal within-endpoint collisions; a T/I-distinct realization rules out universal T/I equivalence.*

Proof. Equation (4) gives a signed edge bijection in G_M , so Lemma 6.1 gives homometry. Sending e_i to a_i and f_i to b_i defines a map from $\mathbb{Z}^{10}$ that kills every row of M , hence factors through the quotient. The first five columns of M are the reduced incidence matrix of the complete graph on six vertices; the rows for edges $(0, i)$ contain the five standard basis vectors, giving rank five. There are ten columns. Finally a universal collision or rigid equivalence remains true under every homomorphism and therefore under the given realization. $\square$

The rank in (5) belongs to the original signed matching. A map from G_M into a cylinder or a finite cyclic group can impose additional relations and lower its rank; this never changes which branch the original presentation occupies. In particular, we never seek a nonzero map from the finite target C_n to $\mathbb{R}$.

6.1 The exact meaning of a lattice certificate

For row matrices M, N with ten columns, two integer matrices satisfying

$$AM = N, \quad BN = M \quad (6)$$

certify $L_M = L_N$. An equality of rational row spaces does not certify this claim. The latter equality would identify only the free solutions and could identify different torsion groups. We use primitive rational rows only when enumerating height subspaces below.

To certify coordinates for G_M , supply integer matrices U, V, D with

$$UMV = D, \quad \det U, \det V \in \{1, -1\}, \quad (7)$$

where U is 15×15 , V is 10×10 , and D is rectangular diagonal with nonzero entries $d_1, \dots, d_r$ and all other entries zero. The map sending a row vector z to zV induces an isomorphism

$$G_M \cong \bigoplus_{i=1}^r \mathbb{Z}/|d_i|\mathbb{Z} \oplus \mathbb{Z}^{10-r}.$$

Factors of order one are omitted. Thus the coordinates of an old generator are its row of V , reduced modulo $|d_i|$ in each torsion position. All of (6)–(7) can be checked by integer multiplication and determinants. Discovery by Hermite or Smith algorithms is useful; their claimed normal forms are not accepted without these identities. A collision certificate can instead be a row c with $cM = e_i - e_j$ (or $f_i - f_j$), $i \neq j$. A rigid-equivalence certificate consists of the corresponding six integer row identities.

6.2 The finite torsion bound and the finite branch

Lemma 6.3. *The torsion subgroup T_M of the original presentation G_M has order at most 135.*

Proof. First every square minor of a reduced oriented incidence matrix has determinant 0, 1 or -1 . In a chosen square submatrix each row has at most two entries; if some row has at most one nonzero entry, expansion reduces the assertion by induction. Otherwise every row has two opposite entries, so the sum of its columns is zero and its determinant vanishes. Permuting or changing signs of rows preserves this property. Both five-column blocks of M have it.

An $r \times r$ minor using c columns of the first block has a Laplace expansion with $\binom{r}{c}$ summands, each a product of two incidence minors of absolute value at most one. For $r \leq 9$ this is at most $\binom{9}{4} = 126$. If $r = 10$, all ten columns occur. The chosen ten rows correspond to ten distinct edges of the first K_6 . A five-row first-block determinant is nonzero precisely for a spanning tree: a cycle gives dependence; five acyclic edges on six vertices form a connected tree; and leaf elimination gives determinant ± 1 for a tree. The complementary second-block determinant still has absolute value at most one. Therefore

$$|\det M_R| \leq \tau(K_6[R]), \quad |R| = 10,$$

where τ counts spanning trees. No assertion that these summands have a common sign is needed.

The bound on τ is an explicit small finite obligation with no symmetry reduction. Enumerate every $R \subset \mathcal{E}$ of size ten and every $J \subset \mathcal{E}$ of size five. Check connectedness and acyclicity of the graph with edges J , and for every tree count its containment $J \subset R$. Independently, form the 6×6 graph Laplacian of R , remove one fixed row and column, and compute the exact 5×5 determinant. The two answers agree because, for the reduced incidence matrix B_R , Cauchy–Binet gives

$$\det(B_R^T B_R) = \sum_{\substack{J \subset R \\ |J|=5}} (\det B_J)^2 = \tau(K_6[R]).$$

This also proves the Laplacian method's counting interpretation, rather than using an unproved counting shortcut.

The acceptance predicate requires every one of the $\binom{15}{10} = 3003$ labelled R and every one of the $\binom{15}{5} = 3003$ possible J , and equality of the two counts for each R . The complete histogram is:

τ	0	75	96	99	100	104	111
Number of R	6	300	90	360	90	360	360
τ	114	115	120	121	128	130	135
Number of R	360	360	180	72	45	360	60

The counts sum to 3003, and the maximum is 135. The project and the independent review reproduce the full histogram by the two methods. The incidence argument together with this exact finite obligation bounds every nonzero maximal-rank minor by 135.

Finally the ideal generated by the rank- r minors is unchanged by unimodular row and column operations, by Cauchy–Binet applied to those operations and their integer inverses. In a diagonal presentation it is generated by $\prod_{i=1}^r d_i$, whose absolute value is $|T_M|$. Thus $|T_M|$ is the positive gcd of the nonzero maximal-rank minors. It is no greater than any one of their absolute values and therefore at most 135. $\square$

Corollary 6.4. *If $d = 0$ and $G_M \rightarrow C_n$ realizes six-element endpoints, then, after their independent anchors are removed, both endpoints lie in a cyclic subgroup of order q with $6 \leq q \leq 135$ and $q \mid n$. They are an ordinary subgroup inflation of two homometric six-subsets of C_q .*

Proof. Now $G_M = T_M$ is finite. Its image H in C_n is a cyclic subgroup, so its order q divides n and $|T_M|$, and hence is at most 135. Six distinct endpoint coordinates force $q \geq 6$. Identify H with C_q by the injective map $t \mapsto (n/q)t$. The inverse on H transports the signed matching and the six-element endpoints to C_q . Restoring the separate translations of the original endpoints does not change their vertices in the grammar graph. $\square$

6.3 The real solution space and a fixed congruence

Let

$$V_M = \text{Hom}(G_M, \mathbb{R}) = \{v \in \mathbb{R}^{10} : Mv = 0\}.$$

This real vector space has dimension d . Its points are the coordinates of real homometric six-atom lists, even when distinct universal vertices have identical real coordinates.

Lemma 6.5 (Finite unions of subspaces). *Over an infinite field a finite-dimensional vector space cannot be covered by finitely many proper linear subspaces.*

Proof. For each proposed proper subspace choose a nonzero linear functional vanishing on it. Their product is a nonzero polynomial in a basis of the whole space. A nonzero polynomial cannot vanish at all points over an infinite field: in one variable it has only finitely many roots; induction on the number of variables applies this fact to its coefficient polynomials. The product therefore cannot vanish throughout the space. $\square$

For a permutation π of six labels and $\eta \in \{1, -1\}$, define

$$T_{\pi, \eta} = \{v \in \mathbb{R}^{10} : b_i = \eta(a_{\pi(i)} - a_{\pi(0)}) \text{ for every } i\}.$$

These are linear subspaces. Since $a_0 = b_0 = 0$, every real translation or reflection equivalence of the two multisets has exactly this form for some π, η . Labels with equal coordinates can be matched arbitrarily; there are still only $2 \cdot 6!$ choices.

Lemma 6.6 (Fixed free-quotient alignment). *If the two real multisets are congruent for every $v \in V_M$, then one fixed $T_{\pi,\eta}$ contains V_M . After relabelling, reflecting and reanchoring the second universal configuration, every real solution satisfies $b_i = a_i$ for all i . These changes are invertible integral changes of generators and preserve G_M and d .*

Proof. The hypothesis says that the finitely many subspaces $V_M \cap T_{\pi,\eta}$ cover V_M . Lemma 6.5 implies that one is the entire space. Relabelling and reflection give $b'_i = \eta(b_{\pi^{-1}(i)} - b_{\pi^{-1}(0)}) = a_i$ on V_M . A permutation followed by reanchoring transforms the five coordinates of a labelled endpoint by an integer linear map. Undoing the permutation and reanchoring is its integer inverse. Reflection multiplies that block by -1 . Hence the full ten-coordinate transformation is unimodular. $\square$

The conclusion identifies the free projections, not the group elements E_i, F_i . In a finitely generated abelian group an element is killed by every real homomorphism precisely when it is torsion: write the group as $\mathbb{Z}^d \oplus T$ and use its d coordinate maps. Consequently the aligned $F_i - E_i$ may be nonzero torsion. Imposing $F_i = E_i$ as *integral* relations would discard the cyclic phenomena under investigation.

Substitute $b_i = a_i$ only in the real equations. With six unanchored heights, the resulting rows have the form

$$(u_j - u_i) - \epsilon(u_l - u_k), \quad i < j, k < l, \quad (8)$$

where $u_0, \dots, u_5$ denote the six standard vectors. They have coordinate sum zero. Let S be their rational span. Removing the common translation by setting $a_0 = 0$ gives a five-dimensional height space, and

$$\dim_{\mathbb{Q}} S = 5 - d. \quad (9)$$

Indeed every vector in V_M lies on the aligned graph. Conversely any anchored height vector satisfying (8), placed on that graph, satisfies the original real matching equations and lies in V_M . Thus the anchored kernel of S is exactly V_M .

7 The weighted real-line input and the Bloom branch

This section states explicitly the mathematical formula checked by the line solvers. The result is the previously reviewed I/Iw component, tagged **[PROVED], in-house, computer-assisted**. Its completeness remains conditional on the saved exact UNSAT results and their audited encodings; the elementary interface proof below does not remove that dependency.

Lemma 7.1 (Weighted six-atom line theorem Iw). *Two noncongruent homometric real multisets of total multiplicity six have, after independent translations and reflections and possible interchange, one of the following weakly ordered forms:*

$$\begin{aligned} A_1 &= (0, q - 2p, q, 2q - 3p, 3q - 2p, 3q - p), \\ B_1 &= (0, q - p, q + p, 2q + p, 3q - 2p, 3q - p), \end{aligned} \quad p > 0, \quad q \geq 3p; \quad (10)$$

$$\begin{aligned} A_2 &= (0, q - p, q + p, 3q - 2p, 3q - p, 2q + p), \\ B_2 &= (0, p, 2q - p, q + 2p, 3q - p, 2q + p), \end{aligned} \quad p > 0, \quad 3p/2 \leq q < 2p. \quad (11)$$

For six-element sets the two lower bounds are strict. For integer coordinates the recovered p, q are integers. Every displayed pair is homometric and noncongruent.

Mathematical and computational dependency. The normalization, exact occurrence-count encoding, coefficient-plane exclusions and solver obligation are given in Section 7.1. The saved exact UNSAT results exclude every normalized pair outside the two displayed planes over the unbounded real domain. Their weak-order inequalities give precisely the parameter chambers above; the strict-order version gives the strict lower bounds. Integer parameters are recovered from $p = A_5 - A_4, q = A_2$ in the first form and $p = A_4 - A_3, q = A_1 + p$ in the second. The corresponding gap lists also prove noncongruence. This is the reviewed Iw dependency, with its exact solver trust retained, rather than a new solver run or a bounded-data inference. $\square$

7.1 The exact weighted real-line obligation

The following exposition states the complete mathematical formula whose unsatisfiability is used in BF. It is included so that the real-domain claim can be evaluated independently of a program's normal-form names. A six-atom multiset is a list of six real positions with repetitions allowed; the fifteen unordered pairs refer to distinct labelled atoms, even when their positions coincide.

Lemma 7.2 (Diameter normalization). *Every noncongruent homometric pair of six-atom real multisets has positive common diameter L . After independent translations, reflections and division by L , followed if necessary by interchange of endpoints, it has the form*

$$A = (0, a_1, a_2, a_3, r, 1), \quad B = (0, b_1, b_2, b_3, r, 1),$$

with weakly increasing coordinates, $r \geq 1 - a_1, r \geq 1 - b_1$, and $(a_1, a_2, a_3) <_{\text{lex}} (b_1, b_2, b_3)$. The two lists are not related by reflection about $1/2$.

Proof. The largest labelled pair distance is the diameter, so homometry gives the same diameter for both lists. Diameter zero would give two congruent lists with all six atoms at one point. Anchor the minima at zero and the maxima at one. Remove one labelled endpoint-edge occurrence of length one from the common multiset of distances. In either list the largest remaining value is $\max(a_4, 1 - a_1)$: both those endpoint distances remain, and every other edge is no larger than one of them. This argument still holds if an endpoint is repeated or the remaining maximum is also one. Reflect each list independently to put its shared remaining maximum at coordinate four. The two lists are unequal, since equality would be a congruence; lexicographic interchange is therefore available. Reflection is excluded by the original noncongruence. $\square$

Put $z = (a_1, a_2, a_3, r, 1, b_1, b_2, b_3, r, 1)$. The counterexample formula $\Phi(z)$ is the conjunction of the following predicates:

1. Each displayed list is weakly increasing and the two inequalities $r \geq 1 - a_1, r \geq 1 - b_1$ hold.
2. The strict lexicographic inequality of Lemma 7.2 holds, and $\neg \bigwedge_{i=1}^4 (A_i = 1 - B_{5-i})$ holds.
3. With $d_{ij} = A_j - A_i$ and $e_{kl} = B_l - B_k$, for every labelled $i < j$,

$$\sum_{k < l} [d_{kl} = d_{ij}] = \sum_{k < l} [e_{kl} = d_{ij}].$$

Here $[P]$ is one if P holds and zero otherwise.

4. The vector z belongs to neither of the two linear planes given by the two height lists displayed in BF.

The indices in the last condition run through the ten coordinates after omitting the two anchored zeros. If a template has coefficient pairs (u_k, v_k) , choose i, j with $\Delta = u_i v_j - v_i u_j \neq 0$. Membership in its plane is the conjunction, for all ten indices k , of

$$\Delta z_k = (u_k v_j - v_k u_j) z_i + (-u_k v_i + v_k u_i) z_j.$$

Thus nonmembership is the negation of a specified finite conjunction; there is no bounded parameter grid or rationality condition.

Lemma 7.3 (Encoding equivalence). *Φ is satisfiable over the reals if and only if there is a normalized noncongruent homometric six-atom pair outside both displayed planes.*

Proof. Only the distance-count and plane predicates need explanation. The distance equations give equality of multiplicity for every value occurring among the fifteen A edges, including zero. The required multiplicities sum to fifteen, so there is no room for a further B value. Conversely, equal multisets satisfy every equation. Equalities among labelled atoms are retained; the directed autocorrelation adds six diagonal zeros and twice the unordered zero distances to both sides. Cramer’s rule shows that the displayed plane equations are exactly the existence of real parameters p, q with $z_k = u_k p + v_k q$. The other predicates are the normalization and exclusion conditions already proved. $\square$

Finite solver obligation U. The exact formula Φ is unsatisfiable. This is the saved exact linear-real-arithmetic obligation checked by the weighted builder and a separate consecutive-gap/occurrence-bijection encoding. Its truth at all real positions is a solver dependency of this computer-assisted proof. A bounded integer recount does not establish U. Saved Z3 and cvc5 results, input hashes and internal proof-checking settings are specified in the evidence inventory; their proof objects have not been translated into an independent general-purpose proof assistant.

Accepting U, Lemmas 7.2 and 7.3 put every pair in one of the two planes. Recovering the parameters in the first plane gives $p = A_5 - A_4$, $q = A_2$. The weak consecutive gaps force $p \geq 0$ and $q \geq 3p$; $p = 0$ makes the lists equal. In the second plane, $p = A_4 - A_3$, $q = A_1 + p$; the gaps force $p > 0$, $3p/2 \leq q \leq 2p$, and the upper endpoint makes the lists equal. Thus the exact domains are the relaxed chambers stated in BF. Their formal positive-distance identities prove soundness even on the allowed repeated-coordinate boundaries. For completeness their consecutive gap lists are, respectively,

$$\begin{aligned} A_1 &: (q - 2p, 2p, q - 3p, q + p, p), \\ B_1 &: (q - p, 2p, q, q - 3p, p), \\ A_2 &: (q - p, 2p, 2q - 3p, p, 2p - q), \\ B_2 &: (p, 2q - 2p, 3p - q, 2q - 3p, 2p - q). \end{aligned}$$

The first coordinates after zero in the first pair differ by $p > 0$. Reflecting A_1 makes that coordinate p , whereas B_1 has $q - p \geq 2p$. In the second pair the two coordinates are $q - p$ and p , which differ because $q < 2p$; the reflected A_2 has coordinate $2p - q \leq p/2 < p$. Anchored equal-diameter lists can be congruent only by equality or this reflection. Thus both forms are noncongruent, including their allowed repeated-coordinate boundaries. Integer input coordinates recover integer p, q after undoing the diameter normalization: the formulas use only integer point differences. For distinct six-sets the lower bounds become strict. This is the full logical passage from U to Iw and I.

Lemma 7.4 (BF: noncongruent real projection). *Let A, B be homometric six-element subsets of an abelian group K . If some $\ell : K \rightarrow \mathbb{R}$ sends their six-atom lists to noncongruent multisets, then, up to independent rigid motions and interchange, A, B have the group-valued Bloom formula*

$$X = \{0, p, q - 2p, 2q - 2p, 2q, 3q - p\}, \quad Y = \{0, p, q + 2p, 2q - p, 2q + p, 3q - p\}, \quad p, q \in K.$$

The conclusion depends on Iw and the finite integral certificate predicate described in this proof.

Proof. Use Iw to normalize the two projected lists. The needed translations subtract actual elements of A or B , and reflections use inversion and such translations, so they can be performed in K . Label equal height atoms arbitrarily. The resulting projected coordinates have one of (10)–(11). At this stage their p, q are real numbers; they are not assumed to lift to K . The existence of group-valued parameters is what must be proved.

For each form compute the fifteen coefficient differences of its ordered list. Both sides have the same fifteen distinct coefficient pairs. These can be checked by subtracting entries of the displayed lists; as real expressions the two common multisets are

$$\begin{aligned} \mathcal{D}_1 = \{ & q - 3p, 2q - 3p, q - 2p, 2q - 2p, 3q - 2p, \\ & q - p, 2q - p, 3q - p, q, 2q, p, q + p, 2q + p, 2p, q + 2p \}, \\ \mathcal{D}_2 = \{ & 2q - 3p, 2q - 2p, 3q - 2p, q - p, 2q - p, 3q - p, \\ & q, 2q, p, q + p, 2q + p, 2p - q, 2p, q + 2p, 3p - q \}. \end{aligned}$$

Put $t = q/p$. Two unequal coefficient forms $up + vq$ and $u'p + v'q$ can agree only when $v \neq v'$ and $t = (u' - u)/(v - v')$. Enumerate these rational slopes for the $\binom{15}{2}$ pairs and retain the appropriate chamber. This yields precisely 3, 4, 5 for the first form and 3/2, 5/3 for the second. Away from these slopes there is exactly one matching, the formal coefficient matching. All forward real differences are positive there, so their signs are +1.

At an exceptional $t = a/b$, evaluate each coefficient pair at the integer parameters $(p, q) = (b, a)$ and group edges into equal-height bins. In a bin of size k , enumerate all $k!$ bijections. A zero bin also has 2^k independent sign choices; a positive bin has only the positive sign. Thus the exact finite input is:

Form	Generic	First slope	Second slope	Third slope	Total
1	1	128 at 3	16 at 4	4 at 5	149
2	1	128 at 3/2	4 at 5/3	—	133

These 282 signed matchings exhaust all possible lifts of every noncongruent projected pair, because only the five explicitly computed slopes permit additional edge coincidences. This is an argument over all real parameters, not a parameter sample.

For clarity, the complete *acceptance predicate* for this finite step is as follows. Reconstruct every matrix from its signed labels by (4). For each of the seven cases, compare the saved labelled matchings with the full independent bin-permutation or edge-by-edge enumeration, including signs on zero edges; require exact set equality and absence of duplicates. Each matrix has an assigned representative with both integer identities (6). Each representative has a certified diagonalization (7). Finally, in the resulting exact group coordinates require one of:

1. two distinct labels on the same endpoint have identical coordinates;
2. explicit p, q , endpoint interchange if used, translations t_X, t_Y and a sign η satisfy

$$\text{multiset}(E_i) = t_X + \text{multiset}(X(p, q)), \quad \text{multiset}(F_i) = t_Y + \eta \text{multiset}(Y(p, q)),$$

with the sides interchanged when specified. These equalities are checked by evaluating all twelve coefficient formulas; they are not inferred from a successful or failed parameter search.

The saved package passes this predicate. Its 21 distinct labelled integral lattices consist of 8 binary $\mathbb{Z}^2$ presentations, 3 binary $\mathbb{Z}$ presentations, 8 binary $\mathbb{Z} \oplus C_2$ presentations, all with explicit Bloom identities, and 2 $\mathbb{Z}$ presentations with forced collisions. Distinctness of the representative lattices can also be certified by a row whose class is nonzero in the other's certified quotient; it is ancillary to coverage.

An actual signed matching of the group pair supplies $G_M \rightarrow K$. The collision outcomes contradict the six-element hypothesis. Every remaining outcome has a Bloom identity already in G_M ; applying the homomorphism gives that identity in K , with distinctness inherited from A, B . This proves BF, with precisely the Iw and finite-predicate dependencies stated above. $\square$

For example the apparently new torsion boundary configuration

$$\{0, g, 3g, 3g + h, 7g, 8g + h\} / \{0, 2g + h, 4g + h, 7g, 7g + h, 8g + h\}, \quad 2h = 0,$$

is already Bloom: take $p = g + h, q = 3g$ and reflect Y, X , respectively, about $8g + h$. Torsion in a matching presentation alone does not prove that its images are non-shadows.

Corollary 7.5 (Intrinsic free-quotient dichotomy). *For a finitely generated universal presentation, either the endpoint multisets in its full free quotient are congruent, or BF applies and the pair is group-valued Bloom. In the congruent case a single labelled alignment is valid for every real solution.*

Proof. If no fixed vector congruence holds in the free quotient, the equalities for each choice of sign, permutation and anchor can hold after a real functional only on a proper subspace of the dual. There are finitely many proposals, so Lemma 6.5 supplies a functional outside all of them. Its two projected multisets are noncongruent and BF applies. In the other case use Lemma 6.6. $\square$

8 The high-free-rank cover and its direct $L3^*$ form

Lemma 8.1 (HR). *Let M be an original signed matching presentation whose universal endpoints have six distinct points and are T/I distinct. If $d \geq 3$, then, conditional on Iw and the finite certificate predicate below,*

$$G_M \cong C_2 \oplus \mathbb{Z}^3.$$

After relabellings and independent rigid motions its two configurations are exactly

$$C = \{0, p, q, p - q + h\}, \quad A = C \sqcup \{r, r + h\}, \quad B = C \sqcup \{p - r, p - r + h\}, \quad 2h = 0, \quad (12)$$

where p, q, r form a basis of the free quotient and h generates C_2 . Every admissible specialization of this family is directly $L3^$. There are no T/I-distinct six-set universal presentations of rank $d = 4$ or 5.*

Reduction and exhaustive finite contract. For each $v \in V_M$, Iw says that its real multisets are either congruent or a labelled Bloom pair. There are finitely many label permutations, orientations and choices of origin. Each fixed Bloom choice defines a linear subspace of the anchored ten-coordinate space of dimension at most two: all coordinates are fixed integer-linear combinations of its two parameters, and anchoring removes the translations. Thus V_M is covered by finitely many intersections with congruence subspaces and Bloom subspaces. Since $\dim V_M \geq 3$, none of

the Bloom subspaces can contain it. Lemma 6.5 gives one fixed congruence alignment. Apply Lemma 6.6 and retain the original integral presentation; only its real heights have $b_i = a_i$.

By (9), the rational height span S has rank at most two. The primitive directions of all nonzero rows (8), with an overall sign removed, form the following explicit S_6 -invariant set $\mathcal{D}$:

$$\begin{array}{ll} (1, -1, 0, 0, 0, 0) & \text{and its 15 projective permutations,} \\ (2, -1, -1, 0, 0, 0) & \text{and its 60 projective permutations,} \\ (1, 1, -1, -1, 0, 0) & \text{and its 45 projective permutations.} \end{array}$$

To verify that these are all possibilities, subtract two signed incidence roots. Equal supports give zero or a multiple of a root; one shared vertex gives a root or the second shape; disjoint supports give the third shape. The counts are respectively $\binom{6}{2}$, $6\binom{5}{2}$, and $\binom{6}{4}\binom{4}{2}/2$. Hence $|\mathcal{D}| = 120$. Dividing by a coefficient gcd here does not divide any row of M .

Enumerate the zero span, every one-direction span, and every two-direction span of $\mathcal{D}$, identifying rational spans by exact row reduction. This enumeration is exhaustive because a span of rank at most two has a basis chosen from its generating rows. Partition these spans by all 720 permutations of the six coordinates. A simultaneous permutation of the aligned endpoints and reanchoring induces an invertible integral change of the original generators, so one representative of each orbit suffices. The saved enumeration has 4176 spans in 25 orbits.

For each representative S , supply an integer height vector $z = (0, z_1, \dots, z_5)$ with the certificate property

$$v \cdot z = 0 \iff v \in S, \quad v \in \mathcal{D}. \quad (13)$$

Such a vector exists: each $v \notin S$ cuts a proper hyperplane in the anchored rational kernel of S ; Lemma 6.5 over $\mathbb{Q}$ gives a point outside all of them, and clearing denominators makes it integral. The stored vector must be checked against all 120 directions. Its purpose is to record all forced signed-edge equalities, not to approximate an arbitrary real solution.

Enumerate every permutation between the fifteen edge occurrences of z and another copy of z that preserves absolute edge differences. For a positive-length bin the sign is uniquely determined by the two oriented differences. For a zero bin both signs are included independently. If a bin has size k , it contributes $k!$, with an additional factor 2^k only at zero. Reconstruct the full integral matrix for every matching by (4). Every original presentation in this branch occurs in this collection after the justified changes of generators: each of its matching rows restricts to a row in S , so (13) makes that signed equality true at z . Conversely any equality at z represents the same rational forced height relation. Extra presentations with different integral rank may be enumerated, which is harmless; none is omitted.

The acceptance predicate for the finite cover has four parts:

1. Independently reconstruct $\mathcal{D}$ and all rank-at-most-two spans; check that the 25 orbit lists are disjoint and their union is the full 4176-span set. Check (13).
2. For every orbit representative compare the entire saved labelled matching set with an independent edge-by-edge enumeration. Require exact set equality, no repeated record, and the bin-factorial count.
3. Each discarded record must carry either $cM = e_i - e_j$ or $cM = f_i - f_j$ for distinct labels, or a permutation π , sign η and anchor k with six identities

$$c_i M = e_i - \eta f_{\pi(i)} + \eta f_k.$$

Check that π is a bijection and all coefficients are integers. These identities prove collision or rigid equivalence in every image.

4. Every remaining record has matrices satisfying (7) with diagonal $(1, 1, 1, 1, 1, 1, 2, 0, 0, 0)$. In the exact quotient coordinates, evaluate a supplied origin, rigid alignment and p, q, r, h and require all twelve master coordinates (12) to equal the universal endpoints. Require $h \neq 0$, $2h = 0$, and determinant ± 1 for the 3×3 free-coordinate matrix of p, q, r .

The last determinant and the order of h prove that these parameters generate the whole $C_2 \oplus \mathbb{Z}^3$, rather than a proper sublattice: their free components give a basis, and h generates the remaining torsion. Thus the record certifies the universal family itself.

The saved files pass this predicate, with 14475 total matchings: 13916 collision records, 555 universal rigid-equivalence records, and four master-family records. The discard records contain 17246 integer row identities. All four survivors have the same height orbit, with span generated by

$$z_0 - z_1 - z_2 + z_5 = 0, \quad z_3 - z_4 = 0;$$

the generic stored vector is $(0, 8, 1, 3, 3, 9)$ and that orbit has 128 matchings. Independent replay checks every record; the numerical totals alone are not the acceptance predicate. The preceding exhaustiveness argument and these identities prove the structural part of HR. Since the survivors have free rank three, none has rank four or five. $\square$

Direct algebraic L3 identity.* In the group ring of any abelian target with $2h = 0$, put $P = x^p$, $Q = x^q$, $R = x^r$, $H = x^h$. Then $H^{-1} = H$ and

$$C = 1 + P + Q + PQ^{-1}H, \quad D = R(1 + H).$$

Direct expansion gives

$$C - PC^* = (1 - H)(Q - PQ^{-1}), \quad (C - PC^*)(1 + H) = 0. \quad (14)$$

Take the fixed block to be C , the moving dyad to be D , and $s = p - 2r$. Its cross distribution is $E = CD^*$, so

$$\begin{aligned} x^{-s}E &= P^{-1}R^2CR^{-1}(1 + H) \\ &= RP^{-1}C(1 + H) = RC^*(1 + H) = E^*. \end{aligned}$$

This is exactly the L3* hypothesis. The shifted dyad is

$$x^sD = PR^{-1}(1 + H), \quad \text{with points } \{p - r, p - r + h\},$$

so its endpoint is precisely B . Six-point distinctness supplies the required disjointness. This proves homometry and the named direct move without a global reflection or a choice of half of s .

For comparison the earlier L5 description is also valid: globally reflect B by $t \mapsto p - t$ to obtain $D + PC^*$, with D fixed and C moving. Equation (14) gives $DC^* = P^{-1}DC$, the strict L5 first-cross-term condition. The direct L3* description is the shorter form of the same family. In a cyclic target six-point distinctness forces $h \neq 0$, hence the modulus is even and $h = n/2$. Its free parameters may otherwise specialize arbitrarily, subject to endpoint distinctness and T/I distinction. $\square$

The BF and HR finite packages are respectively `results/2026-09-30-six-bloom-cylinders.json` and `results/2026-09-30-six-free-rank/`. The checker `tests/test_six_cylinder_branches_review.py` independently rebuilds the signed matching sets and verifies the integer identities, without importing the builders or Smith/Hermite routines. The separate Iw sources, formulas and solver evidence remain `notes/2026-09-30-six-integer-theorem.md` and its logged adversarial review. The present exposition does not claim new novelty or formal proof-assistant verification.

8.1 Congruent real projections at low free rank

We now specify the finite verification used when the original signed matching presentation has free rank one or two. All lattices in this subsection are integral lattices in the same labelled $\mathbb{Z}^{10}$. Only the auxiliary height arrangement is a rational arrangement.

Lemma 8.2 (LR). *Let M be a full signed matching matrix and let $d = 10 - \text{rank}_{\mathbb{Q}} M \in \{1, 2\}$. Suppose that one fixed relabelling, reflection and reanchoring makes every real solution satisfy labelled $B = A$. Every cyclic realization having six distinct points on each side and different T/I classes is a specialization of a Bloom, D, L2, L3*, L4, L5 or L7 construction.*

The proof consists of the mathematical reductions below and the stated finite acceptance predicates. The counts describe accepted existing certificate packages; they are not assumptions about a numerical trend.

The height direction arrangement. Put $A = (0, a_1, \dots, a_5)$ after the fixed normalization of B . Before anchoring, let $\alpha_{ij} = e_j - e_i \in \mathbb{Z}^6$, $i \neq j$. Every substituted matching row is a difference of two such oriented edges. Its nonzero primitive direction, considered up to sign, is a permutation of

$$(1, -1, 0, 0, 0, 0), \quad (2, -1, -1, 0, 0, 0), \quad (1, 1, -1, -1, 0, 0).$$

There are respectively 15, 60, and 45 directions. Denote their union by $\mathcal{D}$. Let S be the rational span of the substituted rows. The space of their height solutions is $S^\perp$, modulo the constant vector. The projection from the original real solution space to the anchored A coordinates is an isomorphism: it is injective because $B = A$, and substitution gives its inverse. Consequently $\dim S = 5 - d$, hence $\dim S$ is three or four. Dividing out integer content here describes S ; it does not change any matching lattice.

Lemma 8.3 (Exhaustive height representatives). *Every rank- r span generated by $\mathcal{D}$ occurs, up to a vertex permutation, by adjoining a direction to a representative of rank $r - 1$. For every such span S there is an integral vector h , with $h_0 = 0$, such that $v \cdot h = 0$ for $v \in \mathcal{D}$ exactly when $v \in S$.*

Proof. Choose a basis of S from its generating directions and remove its last member. A permutation taking the remaining span to a representative also takes the removed member into $\mathcal{D}$, since $\mathcal{D}$ is permutation invariant. Induction proves the first assertion, starting at the zero span. For the second, each $v \notin S$ defines a proper rational hyperplane in $S^\perp$. A finite union of proper hyperplanes cannot cover this vector space. Choose a rational vector outside their union, subtract its first coordinate from every coordinate, and clear denominators. All directions have coordinate sum zero, so anchoring does not alter the required evaluations. $\square$

The finite height check starts with all spans of ranks zero, one and two, and verifies the extension induction by exact rational arithmetic. For each saved representative it applies all 720 vertex permutations, checks orbit size and disjointness, and checks every extension of every preceding representative. Every saved orbit must meet that extension list. Independently, a span can be represented by its primitive exterior minors: two full-rank row lists represent the same rational span exactly when their exterior coordinate vectors are proportional. The accepted rank-three table has 43 770 spans in 104 orbits; the rank-four table has 116 401 spans in 211 orbits. Each saved h is checked against all 120 directions using exact dot products and exact span membership. Thus equal heights and equal signed edge heights are precisely the coincidences forced by the representative span, without sampling error.

Roots retain the original integral relations. For one fixed h , orient every nonzero-height edge upward and group these edges by their positive height differences. If a bin has m edges, its compatible bijections are the $m!$ permutations. A full matching respecting h restricts to one element of the product of these bin-permutation sets. Equal-height edges are left for completion. Relabellings inside each equal-height fibre act independently on the source and target lists. They preserve the completion problem and are integral invertible changes of anchored generators. In particular, when the zero-labelled vertex moves, reanchoring subtracts its new coordinate; the inverse relabelling gives an integral inverse.

For an upward source edge (i, j) and target edge (k, l) the root row is $\alpha_{ij}^A - \alpha_{kl}^B$ in anchored coordinates. Upward order need not have $i < j$. To use the fixed $i < j$ convention, multiply the whole row by -1 if the source edge is reversed and transfer that sign to the target; reverse the target edge as necessary and change its sign accordingly. These operations leave its integral row lattice unchanged. No equation $v \cdot h = 0$ is added simply because it holds.

The root acceptance check reconstructs all bins and all within-fibre vertex permutations. For each saved bin permutation p , it constructs its complete orbit $\{\beta p \alpha\}$ under the independent actions, checks the recorded orbit size, and checks disjointness from earlier orbits. Their union must have cardinality $\prod m!$. It reconstructs every saved root matrix from its edge permutation, rather than trusting the saved matrix. Across the 315 height strata this accounts for 4822 240 cross-edge bijections in 328 473 root orbits.

Exact group coordinates and cache equality. A node has a row matrix $H \in \mathbb{Z}^{k \times 10}$ and represents $G_H = \mathbb{Z}^{10} / \text{row}_{\mathbb{Z}}(H)$. Its Smith certificate consists of $U \in \mathbb{Z}^{k \times k}$ and $V \in \mathbb{Z}^{10 \times 10}$ with

$$\det U, \det V \in \{1, -1\}, \quad UHV = D.$$

Here D is a rectangular diagonal matrix with r nonzero entries, where $r = \text{rank } H$. Thus U is not necessarily 15×15 : its dimension changes with the node's actual row count. Keeping the coordinates of rows of V at indices $i < r$ with $|D_{ii}| > 1$, reduced modulo $|D_{ii}|$, and all indices $i \geq r$ gives the ten generator coordinates in $T \oplus \mathbb{Z}^{10-r}$. Entries $|D_{ii}| = 1$ contribute no coordinate. The two anchors are zero. The verifier checks the matrix identity, dimensions, determinants, rank, diagonal data and all twelve recorded point coordinates by integer arithmetic.

The producer caches the row-Hermite matrix of the relation lattice. This cache is local to one fixed height stratum and the same labelled generators; neither rational span equality nor an unrecorded abstract group isomorphism permits a merge. For every root normalization and every parent-plus-branch normalization, the package supplies integral matrices C, C' satisfying

$$CH_{\text{source}} = H_{\text{target}}, \quad C'H_{\text{target}} = H_{\text{source}}.$$

These equalities prove equality of integral row lattices in both directions, including torsion, without trusting a Hermite routine. Canonical minimality is useful for efficiency, not for the coverage proof: two-way lattice equality is the required mathematical contract.

Paired cancellation and exhaustive branching. In G_H form a bin for each unsigned class $\{z, -z\}$ of the fifteen edge differences, retaining its labelled edge occurrences. Give the class the deterministic representative $\min(z, -z)$ in Smith coordinates. Cancel the minimum of the two multiplicities for each common class. This cancels equally many occurrences on the two sides, including order-two classes; it is not cancellation of vertices or of real heights.

If two points on one side coincide in G_H , discard the node as a forced collision. If $B = s + \epsilon A$ in G_H , with $\epsilon \in \{1, -1\}$, discard it as forced T/I. Both conditions persist in every quotient. Otherwise

an empty residual list is a homometric terminal. At a nonterminal choose one residual A edge (i, j) . For each residual B class use its first labelled edge (k, l) and include each sign ϵ satisfying

$$h_j - h_i = \epsilon(h_l - h_k).$$

The child adjoins the actual integral row $\alpha_{ij}^A - \epsilon\alpha_{kl}^B$. Both signs are included if both satisfy this equality, as happens for zero heights.

Lemma 8.4 (Coverage and termination of completion). *Suppose a full presentation G_M in this stratum is a quotient of a node G_H . Unless its realization has a collision or is T/I, it factors through one recorded child of every nonterminal encountered. Every path terminates after at most fifteen uncanceled-edge decreases.*

Proof. The cancelled classes remain equal in G_M . Subtracting the same multiset from the two equal full edge multisets therefore leaves equal residual multisets in G_M . The chosen A occurrence equals some residual B occurrence up to sign there. Replacing that occurrence by the saved representative of its unsigned class changes at most the sign, since equality up to sign already holds in G_H . Both possible signs are tested. The common real height homomorphism is defined on G_M , so the resulting signed height equality is among the branches. The newly adjoined row holds in G_M and gives the required factorization.

Existing unsigned equalities persist on passing to a quotient. The new equality pairs one occurrence from each residual side that had not been paired before. Even if further classes merge, a matching of occurrences extending all previous cancellations and this new pair exists. Hence the maximum cancellable multiplicity increases by at least one, and the residual multiplicity strictly decreases. It starts at most fifteen. Two-way lattice certificates make cache reuse a reuse of the same labelled quotient, so this decrease also precludes cycles after merging. $\square$

This argument uses the original presentation and its real homomorphism; it does not posit a nonzero homomorphism from the finite cyclic target back to $\mathbb{R}$. The root-orbit reduction may first replace M by its within-fibre relabelling; that presentation has the same realization up to the recorded changes of labels and anchors.

Here is a complete abstract version of the completion checker:

```
for each height stratum:
  verify height orbit cover, generic heights, and cross-root orbit cover
  for each root: check both integral inclusions into its saved node
  visit each reachable node once:
    verify Smith coordinates and all point/edge classes
    recompute residual lists by paired multiplicity cancellation
    if collision: verify two distinct labels have equal coordinates
    elif T/I: verify the saved sign and shift give the entire other set
    elif homometric: require empty residuals and six points on both sides
    else:
      require chosen edge belongs to the residual A list
      regenerate every height-compatible (B representative, sign)
      require the saved branch list equals this list, without duplicates
      for each branch: check its exact row and both lattice inclusions
      require the child's residual multiplicity is strictly smaller
  require every saved node is reachable and every root orbit is represented
```

The package must report `complete=true`, no error, and certified normalizations; a capped or interrupted search is rejected. The accepted cover has 10 602 nodes: 7 429 forced T/I, 2 207 collisions, 620 homometric terminals and 346 branching nodes, with 974 branch edges. The 658 894 integral inclusion identities are exactly twice the number of roots plus branch edges.

Terminal mechanism predicates. For every homometric terminal the mechanism atlas must contain exactly one record with the same stratum, node identifier, points and group orders. Write $\tilde{B} = s + \epsilon B$ for its certified alignment. For a block certificate check a nonempty disjoint partition $A = U \sqcup W$, its proposed moved block W' , and $\tilde{B} = U \sqcup W'$. With $P = UW^*$, the exact integer coefficient tests are

label	W'	acceptance identity
L2	$x^t W$	$x^t P = P$
L3*	$x^t W$	$x^{-t} P = P^*$
L4	$x^t W$	$2t = 0, \quad x^t(P + P^*) = P + P^*$
L5	$x^t W^*$	$P = x^{-t} U W.$

The group-ring soundness proofs given above apply literally. A verifier expands these distributions as integer counters; it never accepts a label merely because an endpoint autocorrelation check succeeds.

A Bloom certificate supplies p, q , anchor and alignment. Evaluate the six displayed Bloom formulas and require exact endpoint equality with multiplicity. A D certificate supplies an anchor, four-set C and steps a, b . Check the two endpoint formulas and expand

$$K = C + x^{a+b} C^* + (1 + x^a)(1 + x^b), \quad (1 - x^a)(1 - x^b)K = 0.$$

The cyclic periodic-kernel lemma proved above converts this exact identity to the stated nonnegative periodic-weight D grammar. If specialization makes the two equal-sum dyads share a point, they share both points and the endpoints agree; otherwise the four corners remain distinct.

An L7 certificate supplies a generator h of a subgroup H of order $m \in \{2, 4, 6, 12\}$. Enumerate its m multiples and check their distinctness and $mh = 0$. Form $S = A + H$, require $|S| = 12$, check exactly $m/2$ points of A in every occupied H -coset, and require $\tilde{B} = S \setminus A$. Its previously proved specialization lemma is essential: a surviving non-T/I six-point cyclic image preserves the subgroup order and occupied cosets. Thus the image is a literal L7 construction, not only a transported polynomial identity.

The producer's accepted 611 direct terminal certificates comprise 271 L4, 135 L3*, 114 L7, 53 L5, 16 L2, 14 Bloom and 8 D. A different independent search priority supplies 481 D, 71 L5, 24 L3*, 20 L4, 14 Bloom and one L2, again leaving exactly nine terminals. These distributions are not invariants or a disjoint classification; acceptance depends on the supplied identities.

All cyclic images of noncyclic torsion. For any remaining terminal write $G_H = T \oplus \mathbb{Z}^r$ with $T = \bigoplus_{i=1}^k C_{d_i}$ and $E = \text{lcm}(d_i)$. Enumerate all tuples $j_i \in \{0, \dots, d_i - 1\}$, set $c_i = (E/d_i)j_i$, $g = \text{gcd}(E, c_1, \dots, c_k)$ and $q = E/g$. The character has actual image order q , and its surjective image map is

$$\theta_j(t_1, \dots, t_k) = \frac{\sum_i c_i t_i}{g} \pmod{q}.$$

Retain the free coordinates unchanged. If $q = 1$ omit the torsion coordinate.

Lemma 8.5 (Actual-image character factorization). *Every map $f : T \oplus \mathbb{Z}^r \rightarrow C_n$ factors through one of the displayed quotients $C_q \oplus \mathbb{Z}^r$, with $q \mid n$.*

Proof. The image $f(T)$ is cyclic of order q dividing both E and n . Identify it with C_q , and embed C_q in C_E by multiplying by E/q . Each generator of order d_i then maps to an element killed by d_i , hence to $(E/d_i)j_i$ for one enumerated j_i . Dividing the resulting character by its common divisor $g = E/q$ recovers the chosen map onto C_q . Extend its factor map by the original images of the free generators. These images are unrestricted and need not be disjoint from the torsion image. Requiring C_E itself to embed into C_n would be incorrect; only the actual image C_q is used. $\square$

The checker requires every character tuple exactly once, recomputes c_i, E, g, q and all projected point coordinates, then verifies a collision, a T/I witness or one of the preceding mechanism certificates. Seven residual terminals have $T = C_2 \oplus C_2$: each of their four characters forces a collision. Two have $T = C_2 \oplus C_8$: each has sixteen characters, eight colliding and eight admitting L7 certificates (independently also D). Thus all 60 quotients give 44 collisions and 16 mechanisms. Coverage, specialization and this factorization prove LR.

8.2 The bounded finite-source branch

Lemma 8.6 (FT). *For every $6 \leq q \leq 135$, each homometry family of six-subsets of C_q is connected by the stated grammar.*

Proof with its finite acceptance contract. Enumerate six-subsets in two independent ways. First enumerate every positive gap word $(g_0, \dots, g_5)$ with sum q , keeping precisely the lexicographically least word among its rotations and reversals. Equivalently require g_0 minimal and then test all twelve gap images. Cumulative sums give its anchored point tuple. Gap rotations and reversals are exactly translations and reflections of the cyclic point set, so this visits every T/I class once, including symmetric classes. The second traversal enumerates every $0 < a_1 < \dots < a_5 < q$ and keeps the least sorted tuple among all images anchored at one of its six points, with either sign. Any rigid image containing zero has such an anchor, so this test is complete. Lexicographic comparison of point tuples and their gap words first differs at the same gap, so the two canonical conventions agree.

For the first traversal compute the fifteen distances $\min(a_j - a_i, q - a_j + a_i)$. For the second compute directed membership counts $\#\{x \in A : x + d \in A\}$ for $1 \leq d \leq q/2$, halving the antipodal entry. Expand these counts into the sorted fifteen-distance list. Both lists are exact interval inventories. Group complete inventories, retaining groups of at least two classes. The implementation packs fifteen distance bytes in two words and five point bytes in another; for $q \leq 135$ these encodings are injective. Equality uses both full distance words, not a hash. Two traversals must agree family-for-family and in total class count at every modulus, not merely in aggregate totals.

An additional class-count check uses Burnside's formula. A rotation by t has $c = \gcd(q, t)$ cycles of length $\ell = q/c$ and fixes $\binom{c}{6/\ell}$ six-subsets when $\ell \mid 6$, otherwise zero. A reflection with f fixed points fixes $\sum_{j:(6-j) \text{ even}} \binom{f}{j} \binom{(q-f)/2}{(6-j)/2}$, where $f = 1$ for odd q , and $f = 2$ or 0 according to even or odd reflection parameter for even q . Sum over all $2q$ group elements and divide by $2q$.

Each census checkpoint records **methods**, **summary**, **independent**, **families** and **source_sha** 256. Each family stores its complete **members** and **icv**. Reference regressions check the immutable implementation through 22; additional controls attack the 31–33 and 63–65 boundaries.

For every unordered pair within every family, the mechanism file stores either a direct certificate or an explicit gap entry. Certified pairs and gaps must partition the complete pair set, without duplicates or omissions. A direct record stores canonical **a**, **b** and **label**. Bloom records store **parameters** $[p, q']$; R records store a seed index and multiplier, with its seed-order relation checked.

L7 records store subgroup order h ; unit records store u , with $\gcd(u, q) = 1$, endpoint equality and invariant autocorrelation checked. Block/D records store **certificate** $[\lambda, \epsilon, s, \mu, t]$: μ is a six-bit mask selecting U from the recorded A , while ϵ, s align B . Reconstruct both blocks and verify the preceding coefficient identities. The older **swap** label additionally requires its halved-shift solvability condition; it is therefore contained in L3*. For D reconstruct its equal-sum dyads and check the displayed mixed difference of its twelve-unit weight distribution. The cyclic kernel lemma then supplies the required nonnegative periodic decomposition.

Finally, form an undirected graph on each family's members with its verified direct certificates as edges. Require one connected component, using an independent union-find or exhaustive reachability calculation. For every gap pair save and verify an explicit path of these edges; no direct construction is required for that pair. The accepted tables have 725 132 families and 728 424 unordered pair edges, 728 351 direct certificates and 73 gaps, all connected by verified paths. Moduli 6–11 have no nontrivial family. These exact finite conditions, together with soundness of the generators, prove FT. $\square$

Lemma 8.7 (Inflation). *If $q \mid n$, any generating path of six-subsets of C_q embeds to a generating path in C_n under $x \mapsto (n/q)x$. Distinct T/I classes stay distinct. Consequently FT and the rank-ten finite-source bound settle the original free-rank-zero branch at every modulus.*

Proof. The embedding is injective. It preserves every point distinction, partition, subgroup, periodic-weight distribution and group-ring identity, so it transports Bloom, D, block, coset and rigid-template edges. For a unit u modulo q , choose u' congruent to u modulo q and congruent to 1 modulo every prime dividing n but not q . The moduli in these extra congruences are coprime to q , so CRT supplies u' . Primes dividing q cannot divide u' because they do not divide u ; the other primes were excluded explicitly. Thus u' is a unit modulo n and acts on the embedded subgroup exactly as u . Autocorrelation is zero off that subgroup, hence the multiplier condition also transports.

If two nonempty subsets of the embedded subgroup H satisfy $B = s + \epsilon A$ in C_n , choose one point on each side in this equality. Their difference shows $s \in H$. The same equality is therefore a rigid equivalence inside $H \cong C_q$, proving preservation of vertices. For an original rank-ten presentation G_M the group is finite, with $|G_M| \leq 135$ by the minor bound. Its cyclic image has order $6 \leq q \leq 135$ and divides n . Independently anchored inputs lie inside this subgroup, so FT applies and its path embeds as just proved. Restore the original anchor translations at the endpoints. $\square$

Reproducibility and review boundary. The height tables are in `results/2026-09-30-six-free-rank/`; the compressed cross/root/node certificates are in `results/2026-09-30-six-free-dag-all/`; the terminal atlas is `results/2026-09-30-six-free-mechanisms/atlas.json`. The structural checker is `tests/test_six_free_dag_review.py`; the independent terminal checker is `tests/test_six_low_rank_mechanisms_review.py`. They check different obligations and both are required. The additional height checker is `tests/test_six_free_rank_growth.py`. Finite census and edge checkpoints are respectively in `results/2026-09-30-six-large-census/` and `results/2026-09-30-six-pair-mechanisms/`; the finite bridge audit is `tests/test_six_finite_torsion_review.py`. The supplied independent review of the earlier draft rechecked height counts and consequences but explicitly did not replay the internal LR DAG. Its review must not be cited as such a replay. The separately implemented in-house structural and terminal checks are the certificate evidence for that branch. These finite computational claims remain part of a computer-assisted proof, not a proof-assistant-kernel or novelty claim.

9 Completeness of the generating grammar

Proof of Theorem 2.2. Soundness is Proposition 4.1. T/I-equivalent inputs have a length-zero path. For the converse on distinct vertices choose any compatible universal signed matching for a T/I-distinct homometric input. If $d = 0$, Corollary 6.4 and FT give a finite source path; Lemma 8.7 lifts it. If $d > 0$ and some real projection is noncongruent, BF gives B directly. Otherwise one fixed congruence holds on the whole real solution space, as in Section 6. HR handles $d = 3, 4, 5$, while LR handles $d = 1, 2$. Specialization and Lemma 4.2 retain the named admissible generators. These cases exhaust the original presentation dimensions and both projection branches. $\square$

Definition 9.1 (Integer shadow). *A cyclic Z-pair is an integer shadow if it is the reduction of a homometric pair of six-element integer sets. It is purely cyclic if no such lift exists. A cyclic construction label alone does not determine this distinction.*

10 Integer shadows and the large-prime boundary

Proposition 10.1. *Assuming the line obligation U , a cyclic Z-pair is an integer shadow if and only if representatives are an admissible modular Bloom pair.*

Proof. Apply the distinct-point line theorem I to any integer lifts and reduce its integer rigid motions and recovered integer parameters modulo n . For the converse lift the two modular parameters to any integers p, q . Six distinct residues in each formula force six distinct integer terms. The formal Laurent identity for B is valid in $\mathbb{Z}[x, x^{-1}]$, so those integer sets are homometric. An integer congruence would reduce to a cyclic T/I congruence; the latter has been excluded. This also explains why exhaustive testing of the n^2 modular parameter pairs, with rigid alignments, is an exact shadow test despite unbounded integer lifts. $\square$

Corollary 10.2. *If every prime divisor of n is greater than 131, every cyclic six-point Z-pair is an admissible modular Bloom image, under the same line-solver qualification.*

Proof. Take its universal presentation $G_M = \mathbb{Z}^d \oplus T$ and map $f: G_M \rightarrow C_n$. The torsion image has order dividing both n and $|T| \leq 135$. The smallest prime above 131 is 137, so this image is trivial. The pair factors through the free quotient. Its six vertices in each list remain distinct, since their images in C_n are distinct. Here $d > 0$, because a finite presentation with trivial image cannot contain a six-set.

Choose a basis of the free quotient and any integer representatives $c_1, \dots, c_d$ of its images under f . The integer functional $\lambda(w) = \sum_j c_j w_j$ reduces modulo n to f . Any within-endpoint difference w has nonzero image under f , so $\lambda(w)$ is a nonzero integer. Thus λ is injective on each six-element endpoint, with no genericity assumption. Homomorphic transport gives integer homometric six-sets reducing to the original cyclic pair. Proposition 10.1 finishes. $\square$

This sufficient boundary is not an enumeration formula. In particular, deducing $(p-1)(p-11)/12$ at primes requires a separate proof of parameter admissibility, all identifications between parameter images, and all exceptional stabilizers. That counting problem is not settled merely by the large-prime corollary or by matching finite numerical values.

10.1 What the rigid-template certificates additionally prove

The soundness of an R edge requires only equality of its fifteen unsigned differences modulo its listed q . The stronger claim that a faithful seed image is purely cyclic uses an additional, pattern-restricted statement. We give its exact finite contract to keep these two uses separate.

For each row (q, X, Y) of Table 1, order its six coordinates as printed and exhaust all bijections of $\mathcal{E}$ and all signs compatible with those coordinates modulo q . Compatibility means the row relation (4) evaluates to zero modulo q ; both signs are retained for an involution. The saved arithmetic witnesses must, for every such matching and with no omitted or duplicate label list, supply integer unimodular U, V satisfying

$$UMV = \text{diag}(1, 1, 1, 1, 1, 1, 1, 1, q),$$

with the five additional rows zero. If w is the last column of V and v_{seed} is the ten-coordinate anchored seed vector, also require

$$\gcd(w_1, q) = 1, \quad w_i \equiv w_1(v_{\text{seed}})_i \pmod{q} \quad (1 \leq i \leq 10).$$

The seed's first nonanchored coordinate is one. These predicates are checked by integer multiplication, determinants and congruences. The numbers of compatible matchings, in table order, are 192, 96, 48, 48, 48, 48, 16, 12, 8, 8, 4, 4, 4, totaling 536.

Proposition 10.3 (Rigidity restricted to the seed matching patterns). *Under the preceding finite verification, every realization of a compatible seed matching in any abelian group is the displayed seed multiplied by one element t with $qt = 0$. Every faithful cyclic image of each listed seed is purely cyclic.*

Proof. The unimodular diagonal identity identifies its universal presentation with C_q . All old vertices are multiples of the generator, with coefficients w_i . Replacing the generator by w_1 times itself is invertible modulo q and gives exactly v_{seed} . Thus every realization has the asserted form. The allowed orders in the table are checked by reducing both seed lists to each divisor of q , requiring six distinct coordinates and different T/I classes. By the subgroup equivalence argument in Lemma 8.7, that check is valid in every ambient cyclic group.

If the seed itself had homometric six-point integer lifts, their equal absolute edge lengths would give one of the exhausted signed matchings after reduction modulo q . Its anchored integer coordinates would lie in the kernel of a rank-ten M , hence would all be zero, contradicting six distinct points. Now let a faithful image lie in C_n , with $t = (n/q)u$ and $\gcd(u, q) = 1$. Anchor hypothetical integer lifts at their points mapping to zero. All their coordinates are divisible by n/q . Divide by n/q , then multiply by any integer inverse of u modulo q . These operations preserve integer homometry and distinctness and would give integer lifts of the original seed, a contradiction. $\square$

The proposition concerns the listed signed matching patterns, not all six-point pairs. Proper images can admit additional matching patterns; their pure cyclicity does not follow from the faithful-image argument. In particular, this verifies the 21-EDO benchmark's pure cyclicity without assuming that a cyclic factor flip is an integer-line factorization.

The 21-EDO pair

$$\{0, 1, 3, 7, 10, 15\} / \{0, 1, 4, 7, 14, 16\}$$

is the fifth rigid row and has separate exact non-shadow verification. It also has a complement-conjugated swap, a cyclic integral factor flip and a difference-set exchange explanation. Multiple

explanations can coexist. In particular, a factor flip in $\mathbb{Z}[C_n]$ need not be a homometric factorization over the integer line.

Inflation is the injective subgroup embedding of Lemma 8.7. General homomorphic specialization may impose additional torsion relations. Compositions are paths through intermediate T/I classes. These notions are retained separately even where their parameter images overlap.

11 Certificate inventory and reproducibility

All paths below are relative to the supplied project checkout. The companion `evidence-manifest.json` binds the listed proofs, attacks, scripts and saved certificate artifacts to SHA256 hashes. This draft has no public archival DOI or immutable public artifact URL; those must precede a submission claim of publicly reproducible computations. The review package contains sources, proofs and dated six-point mathematical outputs. It also preserves historical exploratory outputs; only the complete certificate packages named by the proof are completeness dependencies.

Component	Written proof / scope of independent evidence
I/Iw	Integer and weighted real-line solver encodings
BF	282 signed matchings; 21 integral presentations
HR	4176 spans; 25 orbits; 14475 signed matchings
C/C2	3003 ten-edge graphs; two spanning-tree counts
R	536 universal matching certificates; all allowed orders
FT	Two complete six-only enumerations through 135; exact paths
LR	315 strata; full quotient cover and terminal arithmetic
G	Separate whole-argument attack and interface repairs
Driver	Portable certificates; separate implementation review

The logical component documents have prefix `notes/2026-09-30-six-` and respective suffixes `integer-theorem.md`, `bloom-cylinders.md`, `free-rank.md`, `completeness.md`, `templates.md`, `finite-torsion.md`, `free-dag.md` and `generation.md`. The synthesis’s proof map pairs each with its separate attack. The elementary D proof and cyclic kernel parameterization are in `shadow-theorem.md` with `shadow-review.md`.

The LR structural replay checks 658894 integral lattice-inclusion identities, all 974 branch edges, all initial matching orbits and completion at 315 expected strata. Smith witnesses check $UMV = D$ and unimodularity. Two-way row identities certify normalization without trusting a library’s claimed normal form. Its terminal arithmetic checker imports no builder, recognizer, Smith or Hermite routine. The independently chosen mechanism priority gives a different noncanonical label distribution while still covering every terminal.

For clarity, the stronger R rigidity verification is needed for the faithful-image non-shadow statement, not for the soundness of an R edge. The latter follows by checking the fifteen unsigned differences in its printed seed row. The real-line obligation U, the spanning-tree bound, the BF/HR finite covers, the full LR cover and the finite-source graph cover are the actual completeness dependencies. Generator discovery priorities, random controls and larger-modulus numerical trends are not additional completeness assumptions.

Use the pinned Python 3.12/NumPy 2.5 environment. The default Python 3.14 / NumPy 1.26 environment on the original host has a demonstrated array-mutation defect and is not evidence for these computations. A focused replay is:

```
.venv/bin/python tests/test_env.py
.venv/bin/python tests/run_tests.py
```

```
.venv/bin/python tests/test_six_cylinder_branches_review.py
.venv/bin/python tests/test_six_free_rank_growth.py
.venv/bin/python tests/test_six_free_dag_review.py -resume -require-complete
.venv/bin/python tests/test_six_low_rank_mechanisms_review.py
.venv/bin/python tests/test_six_finite_torsion_review.py
```

The saved certificate driver also accepts arbitrary modulus. For example:

```
.venv/bin/python src/six_generate.py 21
0,1,3,7,10,15 0,1,4,7,14,16 -out results/example.json
.venv/bin/python src/six_generate.py -replay results/example.json
```

The first command is shown on two lines for typesetting and is entered as one shell command. The driver is a reviewed implementation of G , not a substitute for its completeness proof. Full regeneration instructions, solver inputs and completion flags are in the component proofs. Long replay jobs have resumable checkpoints bound to input hashes.

12 Musical examples and limitations

The four-note teaching example has interval vector $(1, 1, 1, 1, 1, 1)$. A five-note example fixes $\{0, 4, 8\}$ while translating $\{3, 5\}$ to $\{5, 7\}$; its six directed cross differences fill the odd residues once, so L_2 preserves them. The endpoints have vector $(2, 1, 2, 3, 2, 0)$ and different T/I classes. Adjoining the same pitch to a Z-pair is not a general construction: adding 8 to the displayed four-note pair destroys homometry. Pair counts increase from 3 to 6 to 10 to 15, but this teaching progression is not an induction on cardinality.

In C_{12} all fifteen six-note Z-pairs are explained by classical complementation. An ordinary-piano study and a separate retuned 21-EDO miniature demonstrate how equal inventories can support contrasting musical gestures. Registered integer Bloom passages use actual key positions, not a claim that their reduction modulo twelve has six distinct pitch classes. The narrated lesson and editable scores are explanatory companions, not mathematical proof certificates.

The result’s trust base includes exact integer arithmetic in audited C/Python code, independently implemented traversals and the independently encoded solver results for I_w . The separate synthesis reviewer refreshed solver input/proof hashes but did not rerun every component or externally check the solver’s proof objects. There is no full Lean formalization. These limits are material to the meaning of “computer-assisted proof.”

Two online priority searches read relevant primary sources and located no matching full arbitrary-modulus six-point generating theorem. Access gaps include several potentially overlapping full papers and direct bibliographic database searches. Absence from those searches is not evidence of originality. This draft’s theorem is the project’s in-house result; attribution of any new contribution awaits specialist literature review. Unread historical sources are not used as axioms or cited as verified proofs here.

The next mathematical tasks are to remove redundant generators, describe overlapping parameter images, seek useful unique forms and derive counting formulas. Higher cardinalities remain outside this theorem. The project’s separate all-cardinality census and higher-deck definition issues have no role in the proof and retain their recorded limitations.

Authorship, assistance and sharing status

This working manuscript was prepared within the Homometry Program with AI-assisted research, code, mathematical exposition and separate in-house agent reviews. These reviews are not external peer review. No human author, affiliation or approval is inferred by the provisional title block. Before submission or public claims, a human mathematician should inspect the universal presentation cover, weighted-line encodings, lattice checkers, specialization lemmas and prior-art comparison. This manuscript has not been submitted or published.

References

- [1] J. Rosenblatt, *Phase retrieval*, Communications in Mathematical Physics **95** (1984), 317–343. [doi:10.1007/BF01212402](#). Four-point statement and attribution: Theorem 3.9 and the associated remark.
- [2] W. Q. Erickson and N. B. Jones, *Homometric subsets of $\mathbb{Z}_n$ with cardinality 5: classification and enumeration*, arXiv:2412.08997, version 4, 27 August 2026. [arXiv:2412.08997v4](#). The five-point classification and final-section questions were read; preprint status is distinguished from external review of the present result.
- [3] E. Postpischil and P. Gilbert, *There Are No New Homometric Golomb Ruler Pairs with 12 Marks or Less*, Experimental Mathematics **3**(2) (1994), 147–152. [doi:10.1080/10586458.1994.10504286](#). Classical six-mark family and symbolic matching method; its Golomb-ruler scope is not used to infer repeated-distance or cyclic completeness.
- [4] F. Jedrzejewski and T. Johnson, *The structure of Z-related sets*, in *Mathematics and Computation in Music*, LNCS **7937** (2013), 128–137. [doi:10.1007/978-3-642-39357-0_10](#). Author version read: [arXiv:1304.6608](#). General constructions and finite enumeration.
- [5] J. Mandereau, D. Ghisi, E. Amiot, M. Andreatta and C. Agon, *Discrete phase retrieval in musical structures*, Journal of Mathematics and Music **5**(2) (2011), 99–116. [doi:10.1080/17459737.2011.608820](#). Read through the author preprint; final typeset numbering was not assumed.
- [6] J. Mandereau, D. Ghisi, E. Amiot, M. Andreatta and C. Agon, *Z-relation and homometry in musical distributions*, Journal of Mathematics and Music **5**(2) (2011), 83–98. [doi:10.1080/17459737.2011.608819](#). Quotient transport and complementation were read through the author preprint.
- [7] G. S. Yovanof and S. W. Golomb, *The Polynomial Model in the Study of Counterexamples to S. Piccard’s Theorem*, Ars Combinatoria **48** (1998), 43–63. Family N, p. 47; endpoint product labels are interchanged relative to the displayed factors.
- [8] P. Lemke, S. S. Skiena and W. D. Smith, *Reconstructing Sets From Interpoint Distances*, DIMACS Technical Report 2002-37 (2002). Published in *Discrete and Computational Geometry*, Springer (2003), 597–631. The primary report was read; Table 1 supplies the qualified Hosemann–Bagchi historical attribution.
- [9] C. Callender and R. Hall, *Crystallography and the structure of Z-related sets*, Society for Music Theory annual meeting, Nashville, Tennessee, 7 November 2008, conference handout. Cases 3–4; the Bullough attribution is read through this handout.
- [10] J. Goyette, *The Z-Relation in Theory and Practice*, Ph.D. thesis, University of Rochester (2012). Chapter 3, especially Formulas 3.2 and 3.5; the proposed criteria are explicitly empirical (p. 122, footnote 2).